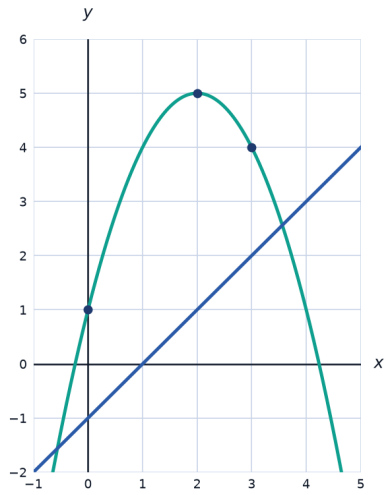
# Graphical solutions to linear equations or inequalities

## Core Concepts: Graphical solutions to linear equations or inequalities

**1** For the model $f(x) = 6x + 9$ in graphical solutions to linear equations or inequalities, evaluate $f(2)$.




**a.** 21

**b.** 27

**c.** 12

**d.** 25

**2** Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

**a.** $x = 1$ and $x = 6$

**b.** $x = 1$ and $x = 5$

**c.** $x = -1$ and $x = -5$

**d.** $x = 6$ and $x = 5$

**3** A data set recorded for graphical solutions to linear equations or inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 6x + 9$ in graphical solutions to linear equations or inequalities, evaluate $f(3)$.

**a.** 27

**b.** 33

**c.** 18

**d.** 31

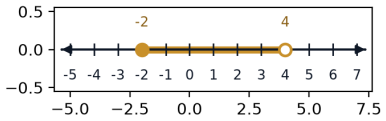
**5** Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

**a.** $x = 2$ and $x = 7$

**b.** $x = 2$ and $x = 6$

**c.** $x = -2$ and $x = -6$

**d.** $x = 8$ and $x = 12$




**6** A data set recorded for graphical solutions to linear equations or inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

## Graphical & Geometric Interpretation

**7** For the model $f(x) = 6x + 9$ in graphical solutions to linear equations or inequalities, evaluate $f(4)$.

**a.** 33

**b.** 39

**c.** 24

**d.** 37

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**8** Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

**a.** $x = 3$ and $x = 8$

**b.** $x = 3$ and $x = 7$

**c.** $x = -3$ and $x = -7$

**d.** $x = 10$ and $x = 21$

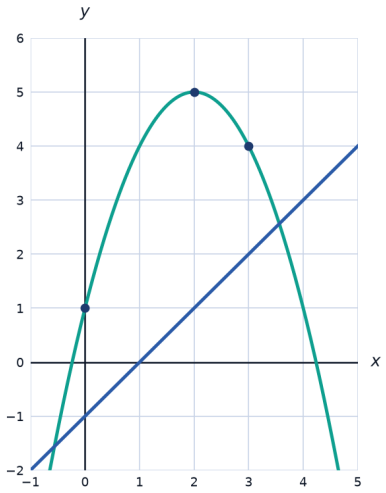
**9** A data set recorded for graphical solutions to linear equations or inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5




## Exam-Style Problem Solving & Modeling

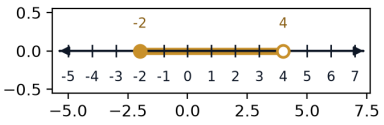
**10** For the model $f(x) = 6x + 9$ in graphical solutions to linear equations or inequalities, evaluate $f(5)$.

**a.** 39

**b.** 45

**c.** 30

**d.** 43




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**11** Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

**a.** $x = 4$ and $x = 9$

**b.** $x = 4$ and $x = 8$

**c.** $x = -4$ and $x = -8$

**d.** $x = 12$ and $x = 32$

**12** A data set recorded for graphical solutions to linear equations or inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

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